



Proof of rotational superradiance

The phenomenon was observed in an experiment where water waves scattered off a vortex with more energy than before <http://digit.in/blackholew>



Clooney sells Tequila company

Smirnoff and Johnnie Walker owner Diageo buys Casamigos, the tequila company co-founded by George Clooney for \$18 <http://digit.in/Clooneyed>

but you would never see a shattered teacup join itself together.

HOW TO TIME TRAVEL

Let's say you're not satisfied with moving just 0.02 seconds into the future like Sergei, and you want to travel 500 or maybe 5000 years into the future. How do you do it?

Recall that time is relative, the faster you travel in space, the slower time ticks for you. So one way into the future is to travel really fast, not rocket ship fast, more like speed of light fast. Sure, we can't travel faster than light,



Time travel is simple. Step 1: Find a really fast spaceship. Step 2: Try not to die

but there's nothing stopping us from getting close to its speed. In order to travel 500 years into the future, you would have to travel at 99.99% the speed of light for 7 years—let's say you do this in a spaceship. Since light travels at 299792458 m/s, your spaceship must travel at 299762478.754 m/s. The fastest rocket ship we've built has a rather modest top speed of 16,260 m/s.

Here we begin to understand the physical limitations of our current technology when it comes to time travel. A problem of speed is really a problem of energy. Theory of relativity also states that mass increases with velocity, so as our spaceship approaches the speed of light, it gets heavier and heavier, and to push it, you must expend more and more energy, the more energy you need, the more fuel you must carry, which adds even more mass—pretty soon the amounts of energy you need to keep it going reach unimaginable levels. Far beyond anything that we're capable of generating. Even if we manage to go nuclear, there

isn't enough radioactive fuel in the world. One candidate could be matter-antimatter reactions, in which 100% of the mass is converted into energy, but the technology just isn't there yet.

As hard as the energy problem may be, it's not the only problem of time travel. Time travel involves time travelers, and human bodies just aren't built to handle the kind of acceleration needed to make this hypothetical 7 year trip work. An average person can withstand about 5 G (49 m/s²) before losing consciousness, if we try to go too fast, our bodies will be torn apart by the centrifugal force. At this rate of acceleration, our time travellers would have to spend about 70 days strapped into their seats, being pushed at their physical limits. In order to overcome this, cryonics and suspended animation could one day be used to put travellers in a state of hibernation in hypersleep chambers, but so far it exists only in fiction.

If you think about it, these are just two big problems that are the most apparent, we haven't even talked about the material science, trip trajectory, on-ship life support, communication systems, and a hundred other things that have to come together to make this work. But speed isn't the only way...

BLACK HOLES & WORMHOLES

Travelling close to the speed of light is one way of slowing down time, the other way is to be in the presence of a strong gravitational field. How strong? The kind that exists near black holes. The strong gravitational field of a black hole bends the space around it, so instead of a straight path, light has to traverse a longer distance along a curvature to get from one point to another. Obviously, a longer would mean lower speed of light, right? No. This is where it gets bizarre: Time will slow down just enough to make sure the speed of light remains constant. The stronger the gravitational field, the higher the curvature, and the greater the time dilation. In fact, if you get to a region within the black hole called singularity where gravity approaches

infinity and spacetime collapses, the lifetime of the entire universe will unfold outside while only a fraction of a second will pass for you. This mind-bending phenomenon of gravity time dilation sounds like a sci-fi trope, but is rigidly enforced everywhere in nature. As seen in the movie *Interstellar*, it is possible to use this time dilation effect for time travel, for every hour that Cooper spent close to the black hole Gargantua, 7 years passed outside its gravitational field, by this math, you would only need to spend 70 hours around the event horizon to experience a 500-year time dilation.

What's the catch? Well, you have to make sure you hover around the event horizon and no closer—if you happen to fall into the black hole's gravity, your body would be torn down into a stream of subatomic particles, like being passed through a microscopic shredding machine, this effect is known as spaghettification. Moreover, the closest black holes we know of are thousands of light years away. It would



An artist's impression of a black hole's singularity sucking in matter from a nebula, it's probably not the brightest idea to get too close to one of these

be easier to do a 7-year time hopping trip than using a black hole. If only there was a shortcut we could take...

Wormholes are hypothetical shortcuts in spacetime that can connect extremely long distances such as billions of light years, different points in time, and even different universes! They're hypothetical, and we have never found one. Scientists like Kip Thorne, who have worked extensively on the science of wormholes, say that one thing we need to ensure is the presence of net